Sitidiyo

# Amashanyarazi

Ibyo umuyagankuba (electrician) akeneye byose ngo ashyire amashanyarazi muri sitidiyo imwe ya kontineri: first fix mbere y'uko inkuta zifungwa, second fix, igerageza (tests) n'icyemezo.

> ***Ubusobanuro:*** *iyi nyandiko yahinduwe na mudasobwa (machine draft). Igomba gusuzumwa n'umuntu uvuga Ikinyarwanda kavukire mbere yo kuyiha abakozi. Iyo hari itandukaniro, kurikiza iy'Icyongereza.*

Status: verisiyo A2, 2026-10-28, PRELIMINARY. Ibipimo byose by'amashanyarazi (ratings) ni ibyo igishushanyo giteganya gusa (design intent), kugeza igihe umuyagankuba (electrician) ufite uruhushya rwa RURA (class A) azategura installation, akayishyiraho, kandi akayiha icyemezo (certificate) hakurikijwe RS 565-1:2011.

Sitidiyo zombi zishyirwamo amashanyarazi kimwe, bityo iyi paji ikora kuri zombi.

English: read this page in English · Gucapa: iyi paji nka PDF · Imizunguruko mu bishushanyo: urupapuro rwa sitidiyo · Igihe bikorwa: gahunda y'imirimo

## Mu ncamake

- **End A (impera A)**: urukuta rufunze rw'impera, iruhande rw'igitanda. **End B (impera B)**: impera ifite inzugi za kontineri, iruhande rw'ubwiherero.
- **Back wall (urukuta rw'inyuma)**: urukuta rurerure rudafite aho gufungukira. **Front wall (urukuta rw'imbere)**: urukuta rurerure rufite urugi runyerera rw'ikirahure.
- Imyanya iri mu mbonerahamwe ya points ni milimetero zibarwa uhereye ku rukuta rurangiye rwa end A (x), uhereye ku rukuta rw'inyuma rurangiye (y), na hejuru uhereye hasi harangiye (z).

| Iki | Igishushanyo (design) |
|---|---|
| Aho umuriro uturuka (supply) | inlet imwe ya 63 A (aho umuriro winjirira), IP67, IEC 60309 y'ubururu (2P+E), hanze ku rukuta rw'inyuma, inyuma neza ya board; isoko y'umuriro iyo ari yo yose ya 230 V |
| Breaker nkuru (main) | breaker (disjoncteur) double-pole ya 63 A (C curve) na DIN kWh check meter (mubazi yo kugenzura); tails za 16 mm² zinyura muri gland |

| Iki | Igishushanyo (design) |
|---|---|
| Board (tableau) | imyanya 24 (modules) mu mirongo ibiri ya 12: breaker nkuru, check meter, SPD type 2, priority relay na contactor yayo, RCBO imwe (30 mA, type A) kuri buri muzunguruko (circuit) |
| Imizunguruko (circuits) | 8, kuva C1 kugeza C8 (imbonerahamwe iri hepfo) |
| Umuriro ntarengwa ukenewe (maximum demand) | 45.3 A (diversified): amatara 0.3 + fans 0.3 + priza (sockets) 16.4 + ikinini muri ibi bibiri: hob 11.6, cyangwa heater ya douche ihita ishyushya amazi 23.9 hamwe na AC 4.3 (D-053) |
| Earthing (guhuza n'ubutaka) | TT: earth rod (piquet de terre) imwe kuri buri kontineri; ntuzigere ukoresha umuriro (supply) uhuza neutral na earth (PME) |
| Inzira y'insinga (cable route) | conduit ya mm 20 mu myanya iri hagati y'ibiti (battens), itwikirwa na spray foam nyuma ya first fix |

| Ryari | Iki | Intambwe |
|---|---|---|
| Nyuma y'ibiti (battens, S03) | First fix: earth rod, conduits, udusanduku (boxes), insinga | S04 |
| Nyuma ya lining n'igikoni | Second fix: board, ibikoresho (devices), igerageza, icyemezo (certificate) | S19 |

Mubazi ya kWh muri buri sitidiyo ni mubazi yo kugenzura (check meter) gusa, igamije gukurikirana: amashanyarazi abarirwa mu giciro cy'icyumba.

## Ingingo zo guhagarara ukagenzura (hold points)

1. **Ibiti bibanza.** Tangira first fix gusa iyo ibiti (S03) bimaze gufatwa kandi impapuro za hasi zimaze gushyirwaho.
2. **Mbere ya foam.** Buri conduit, buri gasanduku na buri nsinga bigomba kuba byashyizwemo, byakorewe igerageza ry'insulation (insulation test), byafotowe urukuta ku rundi, kandi byasinyweho mbere ya spray foam (S05). Foam itwikira conduits: nyuma nta kindi gishobora kongerwamo, keretse ukurura insinga uzinyujije muri conduits.
3. **Mbere yo guha umuriro.** Igerageza ritsinda, kandi icyemezo cya RURA (certificate) gisinywa, mbere y'uko inlet ihabwa umuriro bwa mbere.

# Intambwe

## S04: first fix (imirimo ya mbere)

1. Shyira agasanduku ka board ku rukuta rw'inyuma aho igikoni kiri, n'inlet hanze, inyuma yako neza: ni imyanya DB na PI mu mbonerahamwe ya points.
2. Shinga earth rod (E1) hanze ya end B, ukurure insinga yayo uyigeze kuri earth bar ya board; resistance yayo uzayipima nyuma, mu gihe cy'igerageza.
3. Zamura conduit ya mm 20 uhereye kuri board ujya mu mwanya uri mu gisenge (ceiling void), uyinyuze muri uwo mwanya, hanyuma uyimanure ugororotse ujya kuri buri gasanduku. Nta conduit ihengamye (diagonal); fata buri gasanduku ku giti (batten).
4. Kurura insinga za buri muzunguruko (circuit): 1.5 mm² kuri C1 na C6, 2.5 mm² kuri C2, C3, C7 na C8, 4 mm² kuri C4, 6 mm² kuri C5 (heater ya douche ihita ishyushya amazi).
5. Sohora C6 (amatara yo hanze) unyujije muri gland imwe iri ku gice cy'urukuta rw'imbere kidafite ikirahure (solid front strip), ku ruhande rw'igitanda; hanze, ukomeze muri conduit itangizwa n'izuba (UV-stable).
6. Ubwiherero: itara rya IP65 na fan isohora umwuka (extract fan) ya IPX4 (zone 1, hejuru ya tray kandi munsi ya m 2.25), nta priza, kandi switch yabyo iri hanze y'urugi.
7. Andika nimero (id) ya buri point ku cyuma iruhande rw'agasanduku kayo, kugira ngo umubaji akate impapuro za lining ahantu nyaho.
8. Gerageza continuity n'insulation resistance kuri buri nsinga, ufotore buri rukuta, hanyuma ni bwo usinya uruhushya rwo gushyiramo foam (release).

## S19: second fix (imirimo ya kabiri)

1. Shyiraho ibigize board: breaker nkuru (main breaker) na check meter, SPD, priority relay na contactor, na RCBO zose (B6 kuri C1 na C6, B16 kuri C2, C3, C7 na C8, B20 kuri C4, B25 kuri C5).
2. Shyiraho ibikoresho (devices): amatara, fans, priza, aho hob icomekwa (hob outlet P9: umuriro wayo ucibwa kuri RCBO yayo muri board, igaragara uri kuri hob), n'alarms z'umwotsi n'ubushyuhe (AL1, AL2).
3. Huza (bond) na main earthing terminal: kontineri ubwayo, deck y'imbere, esikariye n'ibyuma byo ku gisenge (roof steelwork).
4. Genzura priority relay: iyo hob ikora, heater ya douche (C5) na AC (C7) bigomba kuzima. Contactor ni iya 40 A: inyuramo 23.9 A za heater.

### Igerageza mbere yo guha umuriro

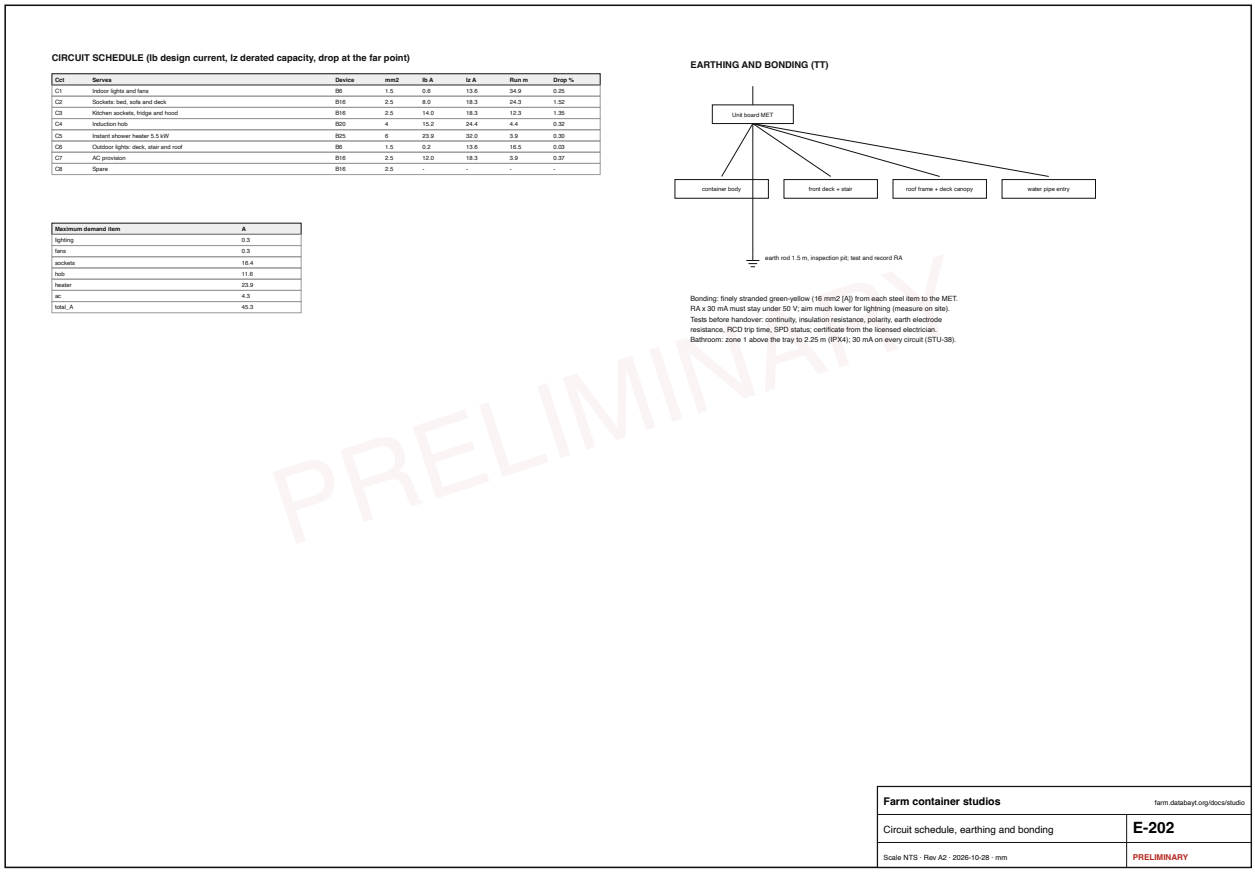
1. Continuity ya buri nsinga irinda (protective conductor).
2. Insulation resistance ya buri muzunguruko.
3. Polarity kuri buri priza no kuri buri tara.
4. Resistance ya earth electrode E1: andika agaciro kabonetse.
5. Igihe buri RCBO imara ngo ihagarike umuriro (trip time).
6. Akadirishya kerekana uko SPD imeze (status window), na priority relay indi nshuro.
7. Icyemezo cya RURA (certificate), mbere y'uko inlet ihabwa umuriro.

## Imizunguruko (circuits)

Each circuit is checked by the model: its current fits the breaker, the breaker fits the cable, and the voltage drop stays under 3 % for lights and 5 % for the rest.

| Circuit | Serves | Breaker | Cable | Design current | Run | Voltage drop | Points |
|---|---|---|---|---|---|---|---|
| C1 | Indoor lights and fans | B6 | 1.5 mm² | 0.6 A | 34.9 m | 0.25 % | L1 L2 L3 L4 L5 L7 S1 S2 S3 F1 F2 |
| C2 | Sockets: bed, sofa and deck | B16 | 2.5 mm² | 8 A | 24.3 m | 1.52 % | P1 P2 P3 P5 |
| C3 | Kitchen sockets, fridge and hood | B16 | 2.5 mm² | 14 A | 12.3 m | 1.35 % | P6 P7 P8 P12 |
| C4 | Induction hob | B20 | 4 mm² | 15.2 A | 4.4 m | 0.32 % | P9 |
| C5 | Instant shower heater 5.5 kW | B25 | 6 mm² | 23.9 A | 3.9 m | 0.3 % | P10 |
| C6 | Outdoor lights: deck, stair and roof | B6 | 1.5 mm² | 0.2 A | 16.5 m | 0.03 % | L8 S5 S4 L10 S6 L9 |

| Circuit | Serves | Breaker | Cable | Design current | Run | Voltage drop | Points |
|---|---|---|---|---|---|---|---|
| C7 | AC provision | B16 | 2.5 mm² | 12 A | 3.9 m | 0.37 % | P11 |
| C8 | Spare | B16 | 2.5 mm² | – | – | – | spare |



CIRCUIT SCHEDULE (Ib design current, Iz derated capacity, drop at the far point)

| Cct | Serves | Device | mm2 | Ib A | Iz A | Run m | Drop % |
|---|---|---|---|---|---|---|---|
| C1 | Indoor lights and fans | B6 | 1.5 | 0.6 | 13.6 | 34.9 | 0.25 |
| C2 | Sockets: bed, sofa and deck | B16 | 2.5 | 8.0 | 18.3 | 24.3 | 1.52 |
| C3 | Kitchen sockets, fridge and hood | B16 | 2.5 | 14.0 | 18.3 | 12.3 | 1.35 |
| C4 | Induction hob | B20 | 4 | 15.2 | 24.4 | 4.4 | 0.32 |
| C5 | Instant shower heater 5.5 kW | B25 | 6 | 23.9 | 32.0 | 3.9 | 0.30 |
| C6 | Outdoor lights: deck, stair and roof | B6 | 1.5 | 0.2 | 13.6 | 16.5 | 0.03 |
| C7 | AC provision | B16 | 2.5 | 12.0 | 18.3 | 3.9 | 0.37 |
| C8 | Spare | B16 | 2.5 | - | - | - | - |

| Maximum demand item | A |
|---|---|
| lighting | 0.3 |
| fans | 0.3 |
| sockets | 16.4 |
| hob | 11.6 |
| heater | 23.9 |
| ac | 4.3 |
| total_A | 45.3 |

EARTHING AND BONDING (TT)



Bonding: finely stranded green-yellow (16 mm2 [A]) from each steel item to the MET.
RA x 30 mA must stay under 50 V; aim much lower for lightning (measure on site).
Tests before handover: continuity, insulation resistance, polarity, earth electrode resistance, RCD trip time, SPD status; certificate from the licensed electrician.
Bathroom: zone 1 above the tray to 2.25 m (IPX4); 30 mA on every circuit (STU-38).

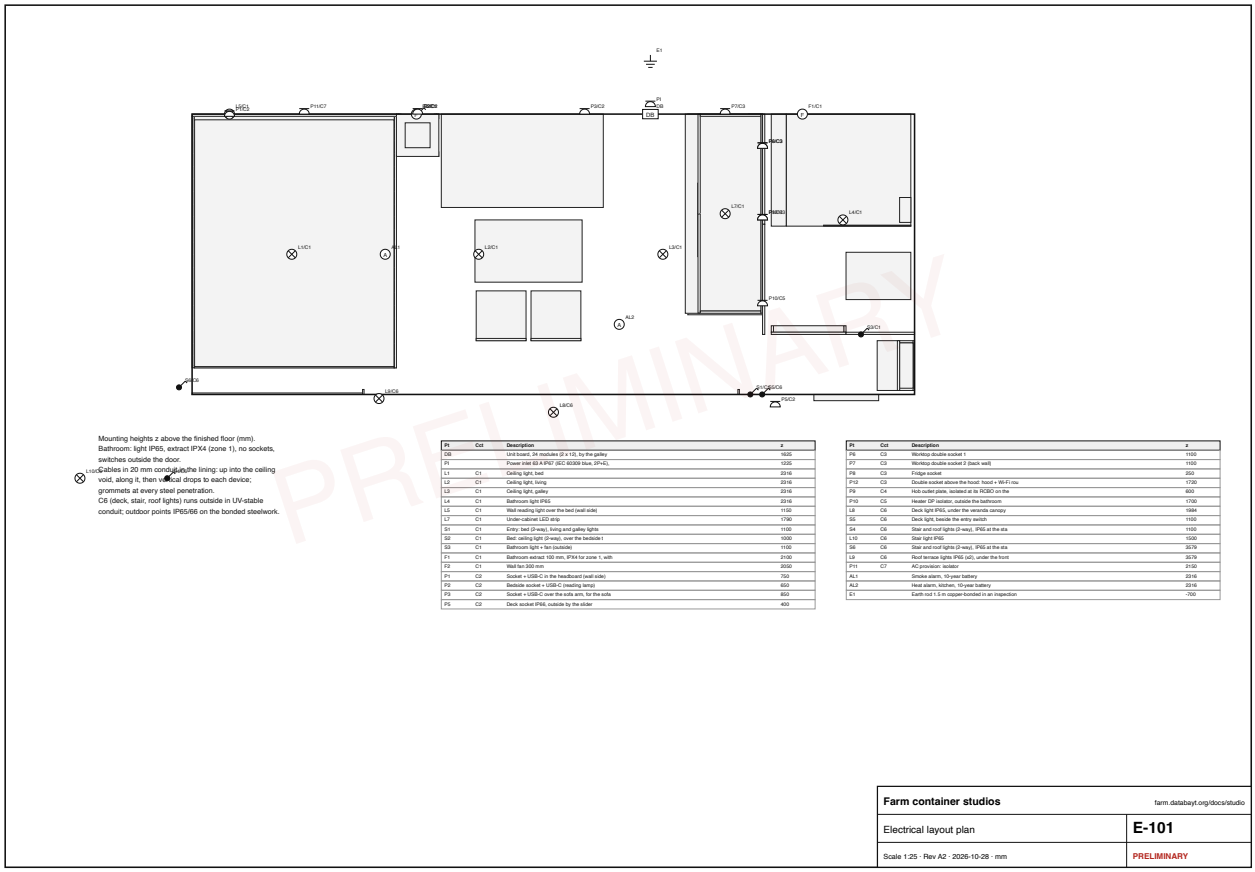
PRELIMINARY

| Farm container studios | farm.databayt.org/docs/studio |
|---|---|
| Circuit schedule, earthing and bonding | E-202 |
| Scale NTS · Rev A2 · 2026-10-28 · mm | PRELIMINARY |

E-202 · Circuit schedule, earthing and bonding · open full size

## Points

33 points on the plan. The bathroom has an IP65 light, an IPX4 extract fan and no sockets, switched from outside the door.




E-101 · Electrical layout plan · open full size

| Point | What | Type | Circuit | Height | Load |
|---|---|---|---|---|---|
| DB | Unit board, 24 modules (2 x 12), by the galley | board | – | 1,625 mm | – |
| PI | Power inlet 63 A IP67 (IEC 60309 blue, 2P+E), outside behind the board | power inlet | – | 1,225 mm | – |
| L1 | Ceiling light, bed | light | C1 | 2,316 mm | 12 W |
| L2 | Ceiling light, living | light | C1 | 2,316 mm | 12 W |
| L3 | Ceiling light, galley | light | C1 | 2,316 mm | 12 W |
| L4 | Bathroom light IP65 | light | C1 | 2,316 mm | 12 W |
| L5 | Wall reading light over the bed (wall side) | light | C1 | 1,150 mm | 4 W |
| L7 | Under-cabinet LED strip | light | C1 | 1,790 mm | 10 W |
| S1 | Entry: bed (2-way), living and galley lights | switch | C1 | 1,100 mm | – |

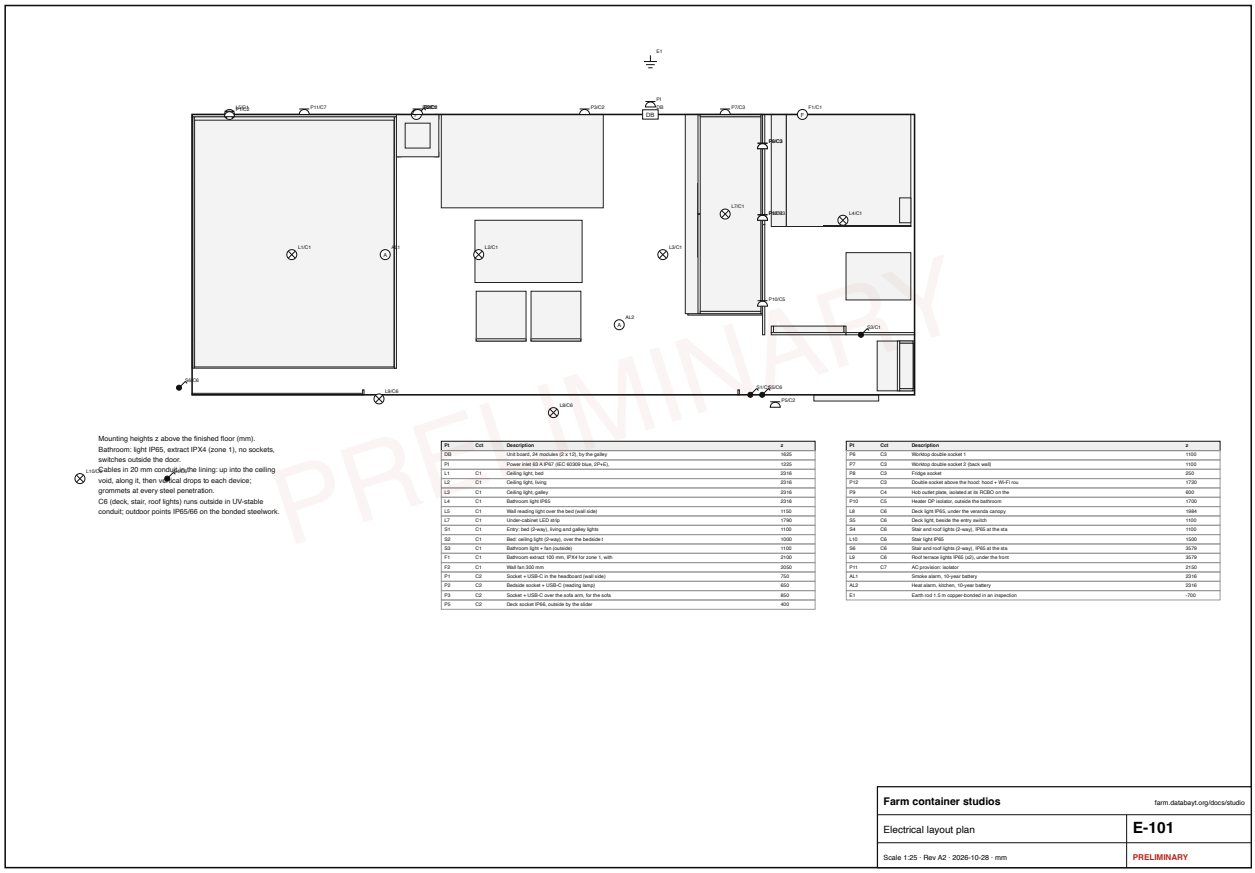
| Point | What | Type | Circuit | Height | Load |
|---|---|---|---|---|---|
| S2 | Bed: ceiling light (2-way), over the bedside table | switch | C1 | 1,000 mm | – |
| S3 | Bathroom light + fan (outside) | switch | C1 | 1,100 mm | – |
| F1 | Bathroom extract 100 mm, IPX4 for zone 1, with run-on | fan | C1 | 2,100 mm | 20 W |
| F2 | Wall fan 300 mm | fan | C1 | 2,050 mm | 50 W |
| P1 | Socket + USB-C in the headboard (wall side) | socket + USB-C | C2 | 750 mm | 200 W |
| P2 | Bedside socket + USB-C (reading lamp) | socket + USB-C | C2 | 650 mm | 200 W |
| P3 | Socket + USB-C over the sofa arm, for the sofa and the table | socket + USB-C | C2 | 850 mm | 500 W |
| P5 | Deck socket IP66, outside by the slider | socket | C2 | 400 mm | 300 W |
| P6 | Worktop double socket 1 | socket | C3 | 1,100 mm | 2,000 W |
| P7 | Worktop double socket 2 (back wall) | socket | C3 | 1,100 mm | 1,000 W |
| P8 | Fridge socket | socket | C3 | 250 mm | 100 W |
| P12 | Double socket above the hood: hood + Wi-Fi router | socket | C3 | 1,720 mm | 120 W |
| P9 | Hob outlet plate, isolated at its RCBO on the board | outlet | C4 | 600 mm | 3,500 W |
| P10 | Heater DP isolator, outside the bathroom | outlet | C5 | 1,700 mm | 5,500 W |
| L8 | Deck light IP65, under the veranda canopy | light | C6 | 1,984 mm | 10 W |
| S5 | Deck light, beside the entry switch | switch | C6 | 1,100 mm | – |
| S4 | Stair and roof lights (2-way), IP65 at the stair foot | switch | C6 | 1,100 mm | – |
| L10 | Stair light IP65 | light | C6 | 1,500 mm | 8 W |
| S6 | Stair and roof lights (2-way), IP65 at the stair head | switch | C6 | 3,579 mm | – |
| L9 | Roof terrace lights IP65 (x2), under the front top rail | light | C6 | 3,579 mm | 20 W |

| Point | What | Type | Circuit | Height | Load |
|---|---|---|---|---|---|
| P11 | AC provision: isolator | outlet | C7 | 2,150 mm | 1,000 W |
| AL1 | Smoke alarm, 10-year battery | alarm | – | 2,316 mm | – |
| AL2 | Heat alarm, kitchen, 10-year battery | alarm | – | 2,316 mm | – |
| E1 | Earth rod 1.5 m copper-bonded in an inspection pit, behind the board | earth rod | – | -700 mm | – |

## Ibibazo bitarasubizwa

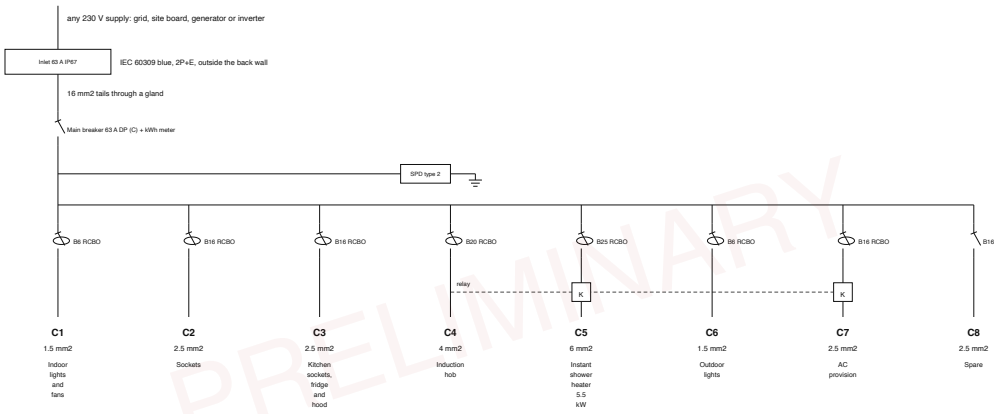
1. Ni iyihe soko izaha inlet umuriro: board yo ku kibanza, generator cyangwa inverter y'imirasire y'izuba? Isoko iyo ari yo yose ya 230 V, 50 Hz irabishobora, niba irinda insinga yayo kuri 63 A cyangwa munsi yaho; paji y'**amashanyarazi** y'ifamu ni yo ibyemeza.

## Ibishushanyo




E-101 Electrical layout plan · open full size

**ONE STUDIO, FROM ITS INLET TO EVERY CIRCUIT**




PRELIMINARY

The studio plugs in: any 230 V source protected at 63 A or less; the 63 A main breaker protects the inlet and tails whatever feeds them.
Priority relay on C4: while the hob runs, contactor K sheds C5 and C7 (heater and AC).
Diversified maximum demand about 45 A (AS-19), within the 63 A inlet. One RCBO 30 mA type A per circuit.
TT earthing: the studio's own earth rod at the board; never rely on the supply's earth (STU-37).
Ratings are design intent: a RURA class A electrician designs and certifies (STU-15), to RS 565-1:2011 (STU-14).

| Farm container studios | farm.databayt.org/docs/studio |
|---|---|
| Single-line diagram: inlet and unit board | **E-201** |
| Scale NTS · Rev A2 · 2026-10-28 · mm | **PRELIMINARY** |

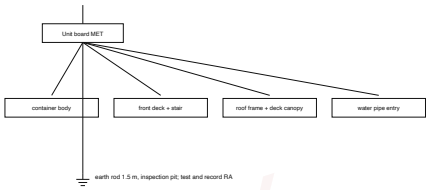
E-201 Single-line diagram: inlet and unit board · open full size

## CIRCUIT SCHEDULE (Ib design current, Iz derated capacity, drop at the far point)

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| C1 | Indoor lights and fans | B6 | 1.5 | 0.6 | 13.6 | 34.9 | 0.25 |
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| C4 | Induction hob | B20 | 4 | 15.2 | 24.4 | 4.4 | 0.32 |
| C5 | Instant shower heater 5.5 kW | B25 | 6 | 23.9 | 32.0 | 3.9 | 0.30 |
| C6 | Outdoor lights: deck, stair and roof | B6 | 1.5 | 0.2 | 13.6 | 16.5 | 0.03 |
| C7 | AC provision | B16 | 2.5 | 12.0 | 18.3 | 3.9 | 0.37 |
| C8 | Spare | B16 | 2.5 | - | - | - | - |

| Maximum demand item | A |
|---|---|
| lighting | 0.3 |
| fans | 0.3 |
| sockets | 16.4 |
| hob | 11.6 |
| heater | 23.9 |
| ac | 4.3 |
| total_A | 45.3 |

## EARTHING AND BONDING (TT)




earth rod 1.5 m, inspection pit; test and record RA

Bonding: finely stranded green-yellow (16 mm2 [Al]) from each steel item to the MET. RA x 30 mA must stay under 50 V; aim much lower for lightning (measure on site). Tests before handover: continuity, insulation resistance, polarity, earth electrode resistance, RCD trip time, SPD status; certificate from the licensed electrician. Bathroom: zone 1 above the tray to 2.25 m (IPX4); 30 mA on every circuit (STU-38).

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E-202 Circuit schedule, earthing and bonding · open full size